

Primate Coexistence, Human Emergence, and the Role of Minimal Coherence ($F_{\min}$) in Speciation

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Abstract

The classical Darwinian narrative describes speciation as a gradual, lineage-based process driven by random mutation and selection. However, the fossil and genetic record of primates reveals a markedly different pattern: long periods of coexistence among multiple fully formed species, punctuated by abrupt disappearances and emergences. This paper argues that primate evolution—and human emergence in particular—is better described as a sequence of boundary-condition-driven phase transitions rather than continuous adaptive optimization. Using the framework of the Theorem of Axiomatic Necessity (TNA), we introduce the concept of a **Minimal Coherence Constraint** ($F_{\min}$) that defines which biological structures are viable under given environmental and physical boundary conditions. Species do not gradually “improve” toward humanity; instead, they persist or vanish depending on their structural compatibility with $F_{\min}$. *Homo sapiens* represents the current configuration compatible with present boundary conditions, not a teleological endpoint. If those conditions change, coherence will re-emerge in different forms, as it has repeatedly throughout evolutionary history.

1. Introduction

Since the late 19th century, human origins have often been framed as a linear evolutionary ascent from primitive primates to modern humans. This framing has motivated the persistent search for a “missing link” — an intermediate, partially formed species bridging ape and human.

Yet over the last decades, paleoanthropology has converged on a different empirical picture:

- Multiple primate and hominin species coexisted simultaneously.
- Newly discovered species appear functionally complete at first appearance.
- Lineages terminate abruptly rather than gradually fading.
- No dense chain of smooth intermediates is observed.

These observations challenge a strictly gradualist interpretation of evolution and motivate a structural re-examination of speciation mechanisms.

2. Empirical Fact: Primate Coexistence, Not Linearity

The African fossil record demonstrates that:

- Australopithecus, Paranthropus, and early Homo species overlapped in time.
- Distinct locomotor, cranial, and dietary strategies existed concurrently.
- Extinction was common, coexistence was the norm.

All primate species did not replace one another sequentially; **they existed together under the same global boundary conditions.**

This directly contradicts the intuitive “ladder” model of evolution and suggests that speciation is not driven by incremental optimization along a single trajectory.

3. The Limits of Gradual Darwinian Computation

Darwinian evolution assumes:

1. Random variation explores configuration space.
2. Selection retains locally advantageous traits.
3. Complexity accumulates gradually over time.

However, this mechanism presupposes **stable boundary conditions.**

When boundary conditions shift abruptly—climate, habitat structure, energy availability—the search space itself changes discontinuously. Under such conditions:

- Random mutation cannot explore the new viable space fast enough.
- Intermediate forms are often non-viable.
- Selection loses its gradual character.

In computational terms, the system lacks sufficient bandwidth to “search” the new space.

4. The TNA Framework and Minimal Coherence ($F_{\min}$)

The Theorem of Axiomatic Necessity (TNA) states that any persistent system requires a non-derivable constraint that maintains coherence.

In biological systems, this constraint manifests as a **Minimal Coherence Condition**:

$F_{\min}$ = minimum structural coherence required for persistence under given boundary conditions

Key properties:

- $F_{\min}$ is imposed by boundary conditions, not by lineage history.
- It is discontinuous under abrupt environmental change.
- It cannot be computed from within the system by gradual variation alone.

Species persist **if and only if** their structure satisfies $F_{\min}$.

5. Speciation as Boundary-Induced Phase Transition

Under the TNA perspective:

- Speciation is not primarily an adaptive trajectory.
- It is a **phase transition** triggered by boundary shifts.
- Multiple species can coexist if they satisfy $F_{\min}$ simultaneously.
- Extinction occurs when a species no longer satisfies the constraint.

This explains:

- Mosaic evolution.
 - Abrupt appearances in the fossil record.
 - The absence of continuous intermediate chains.
 - The repeated failure to identify a single “missing link”.
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6. Human Emergence Reinterpreted

Homo sapiens did not emerge as the “best adapted” primate in a continuous sense.

Rather:

- Environmental fragmentation, climatic variability, and energetic constraints altered boundary conditions.
- A new $F_{\min}$ regime emerged.
- Only certain structural configurations—bipedalism, extended cognition, symbolic communication—remained viable.

Humans are therefore a **boundary-compatible configuration**, not the endpoint of evolution.

7. Relation to the Anthropic Principle

This interpretation aligns naturally with a non-teleological reading of the anthropic principle:

Observers exist only in universes whose boundary conditions allow observer-compatible coherence.

Humans are not privileged outcomes.

They are **current solutions**.

If physical or ecological constants change sufficiently, human coherence fails—and **new coherent structures will emerge**, as they always have.

8. Implications

This framework implies:

- Evolutionary theory is locally valid under stable conditions.
- Speciation under abrupt change requires structural, not stochastic, explanations.
- The “missing link” problem is a category error.
- Human exceptionalism is structurally unsupported.

Evolution describes **optimization within a regime**.

TNA describes **regime transitions themselves**.

9. Conclusion

Primate history does not support a linear Darwinian ladder.

It supports a **coexistence-and-collapse model governed by minimal coherence constraints**.

Humans are not the destination of evolution.

They are its **presently viable configuration**.

When boundary conditions change, coherence will persist—
but not necessarily as us.

Appendix A — Formal Framework: Boundary Conditions, Bifurcations, and Minimal Coherence ($F_{\min}$)

A.1 System Definition

Let a biological system be represented as an open, non-equilibrium dynamical system:

- Internal state space:

$$x(t) \in \mathcal{X}$$

- Boundary conditions (environmental, energetic, physical):

$$B(t) \in \mathcal{B}$$

- Structural constraints (anatomy, metabolism, development):

$$S(x)$$

Persistence of the system requires that the internal dynamics remain confined to a viable subset of state space under the imposed boundary conditions.

A.2 Coherence and Viability

Define **coherence** as the ability of the system to maintain bounded trajectories:

$$\|x(t)\| < \infty \quad \forall t$$

Let $C(x, B)$ be a scalar coherence functional measuring structural compatibility with boundary conditions.

Then viability requires:

$$C(x, B) \geq F_{\min}(B)$$

where:

- $F_{\min}(B)$ is the **Minimal Coherence Constraint**,

- $F_{\min}$ is **externally imposed** by boundary conditions,
 - $F_{\min}$ is **not derivable** from internal system dynamics alone.
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A.3 Evolution as Local Optimization

Under slowly varying boundary conditions:

$$\left| \frac{dB}{dt} \right| \ll \epsilon$$

the system can explore nearby configurations through stochastic variation:

$$x(t + \Delta t) = x(t) + \delta x$$

Selection retains configurations maximizing $C(x, B)$ locally.

This regime corresponds to **classical Darwinian evolution**, which is valid **only when**:

$$\Delta F_{\min} \approx 0$$

A.4 Boundary Discontinuity and Bifurcation

When boundary conditions change discontinuously:

$$\Delta B \rightarrow 0$$

the coherence threshold shifts abruptly:

$$F_{\min}(B_1) \not\approx F_{\min}(B_0)$$

As a consequence:

- Previously viable states become non-viable:

$$C(x, B_1) < F_{\min}(B_1)$$

- Intermediate configurations are typically unstable or lethal.
- Continuous optimization fails.

This defines a **bifurcation event** in the viability landscape.

A.5 Phase Transition in Species Space

Let $\mathcal{X}_v(B)$ denote the viable state manifold:

$$\mathcal{X}_v(B) = \{x \in \mathcal{X} \mid C(x, B) \geq F_{\min}(B)\}$$

Under a boundary shift:

$$\mathcal{X}_v(B_0) \rightarrow \mathcal{X}_v(B_1)$$

This transition is generally **non-homeomorphic**:

- No continuous mapping exists between the two manifolds.
- Trajectories must either **jump** or **terminate**.

This mathematically formalizes **speciation as a phase transition**, not as a gradual deformation.

A.6 Coexistence of Species

Multiple species x_i can coexist if:

$$C(x_i, B) \geq F_{\min}(B) \quad \forall i$$

There is no requirement that species lie on a single lineage or optimization path.

Thus:

- Coexistence is structurally expected.
- Linear ancestry is not required.
- The fossil record should display **parallel viable forms**, not chains.

This matches observed primate and hominin data.

A.7 Extinction as Structural Incompatibility

Extinction occurs when:

$$\forall x \in S_i : \quad C(x, B) < F_{\min}(B)$$

No amount of local optimization can rescue the system if the coherence threshold exceeds its structural capacity.

Extinction is therefore **deterministic under boundary shifts**, not a failure of adaptation.

A.8 Human Emergence

The emergence of the genus *Homo* corresponds to a new viable manifold:

$$\mathcal{X}_v^{\text{Homo}}(B)$$

enabled by:

- altered climatic variability,
- increased energetic demands,
- expanded cognitive and social coherence.

Humans are **solutions to a constraint**, not products of cumulative optimization.

A.9 Formal Statement (Boundary-Driven Speciation Theorem)

Theorem (Boundary-Driven Speciation)

Let a biological system operate under boundary conditions $B(t)$.

If $F_{\min}(B)$ changes discontinuously, then speciation cannot proceed via continuous stochastic optimization. Viable configurations must emerge as discrete structural transitions or not at all.

A.10 Interpretation

- Darwinian evolution is a **local approximation**.
 - Speciation under abrupt change is **constraint-driven**.
 - The “missing link” problem is mathematically ill-posed.
 - Persistence is governed by **minimal coherence**, not gradual advantage.
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Closing Remark (Appendix-safe)

▮ *Evolution optimizes within regimes. Speciation occurs when regimes themselves change.*